Recorded MAX7219 Presentation

Prove every register intent without pretending pixels changed

E0 COPIED POLICY REPLAY — ZERO DISPLAY HARDWARE

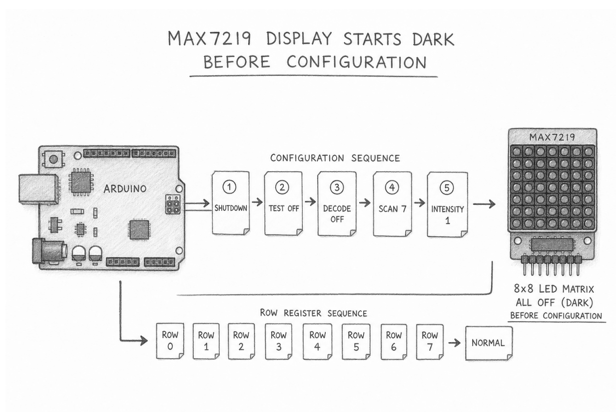
Lesson	059 — component	**Time**	120–180 minutes
Prerequisite	Lesson 058 transaction evidence	**Board**	compile-only Mega replay
Capacity	fixed 8-by-8 logical frame	**Observation**	command and receipt result cells

Responsibility. `Max7219PresentationPolicy` transforms one copied 8-by-8 logical frame into a bounded sequence of MAX7219 register commands. It owns policy state and fault provenance. It owns no SPI bus, chip-select pin, module, resistor, display, supply, clock, retry loop, or powered circuit.

1 Begin dark, then earn normal operation

The MAX7219 is a write-only presentation endpoint in this lesson. A controller can submit bytes, but it cannot read pixels back. Lesson 059 therefore starts with a narrower claim: the policy can prove which command it requested and which copied transport receipt it accepted.

Initialization keeps the logical device dark while it requests configuration. The ordered sequence disables display test, disables decode mode, selects all eight rows, sets bounded intensity, clears rows one through eight, and requests normal operation last. Display test is not used as learner self-test evidence.



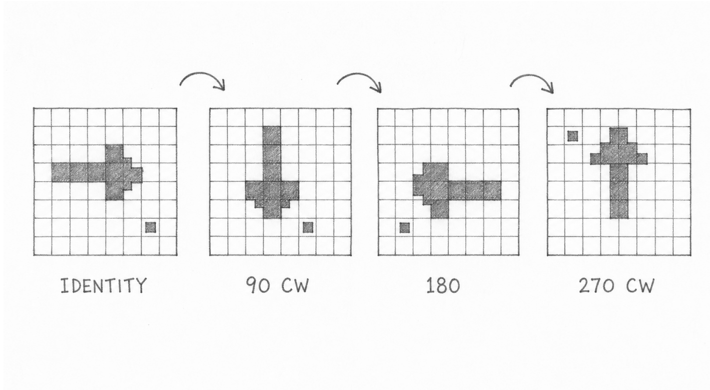
Alternative text: a pencil storyboard begins with a dark eight-by-eight matrix, passes through configuration and eight cleared rows, then reaches normal operation only after the final accepted receipt.

Predict why "normal operation last" matters:

1. If row clearing happened after normal operation, what stale pattern might briefly be presented? ____________________
2. Which copied evidence can prove the request order? ____________________
3. Which physical fact remains unknown at E0? ____________________

2 Separate a logical frame from wire orientation

The independent frame generator owns eight logical row bytes. Logical row zero is the top row, and bit seven is the leftmost pixel before transformation. Orientation is fixed at initialization as `Identity`, `Rotate90`, `Rotate180`, or `Rotate270`. This makes a module-specific rotation explicit instead of silently editing artwork.



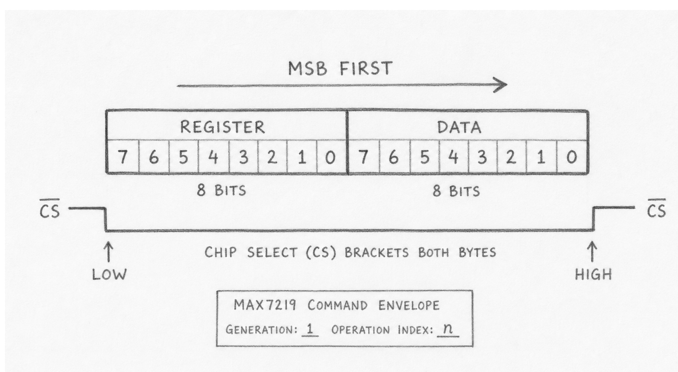
Alternative text: four hand-drawn eight-by-eight grids show one asymmetric arrow transformed through identity and rotations of 90, 180, and 270 degrees; a separate corner marker makes the clockwise movement visible.

Orientation	Predict the logical top-left pixel	Check
`Identity`	remains at physical top left	transformed row 0, bit 7
`Rotate90`	moves to physical top right	expected row and bit
`Rotate180`	moves to physical bottom right	expected row and bit
`Rotate270`	moves to physical bottom left	expected row and bit

Use asymmetric test art. A filled square can look correct after the wrong rotation and therefore proves very little.

3 Read one register envelope

Every published command is a copied value. It carries the owner token, lifecycle generation, configuration revision, presentation generation, operation index, register-address byte, and data byte. The intended MAX7219 transfer is exactly 16 bits, address first and data second, MSB first.



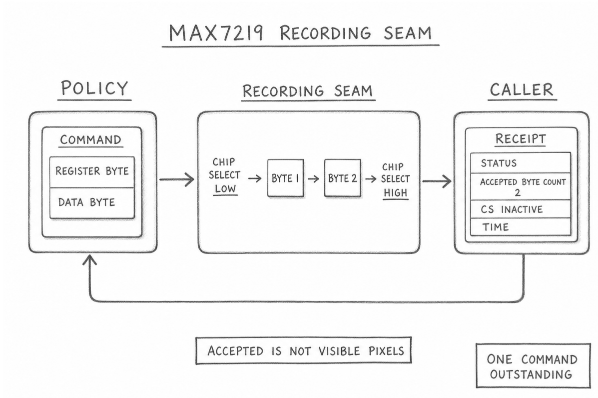
Alternative text: a pencil-drawn envelope contains correlation fields beside two ordered byte cards, first register address and then data, with an MSB-first arrow and a chip-select bracket around both bytes.

Register intent	Address	E0 meaning
decode mode	`0x09`	request no BCD decoding
intensity	`0x0A`	request a validated value from 0 through 15
scan limit	`0x0B`	request all eight rows
shutdown	`0x0C`	request dark or normal operation
display test	`0x0F`	request test off
row 1 through 8	`0x01`–`0x08`	request transformed row data

The canonical replay fixes intensity at 1. That is a logical register value, not proof of safe LED current. A future powered adapter must qualify the exact module, RSET network, supply, current, and thermal behavior.

4 Let the caller pace transport

The policy never invokes SPI code. One service call may publish at most one command. The caller records the two-byte transaction and later returns one copied receipt. That split keeps policy replay deterministic and prevents a hidden retry, wait, callback, or catch-up loop.



Alternative text: a hand-drawn policy box passes one command card to a recording transport box; a separate receipt card returns with the same correlation fields, byte count, chip-select cleanup mark, time, and status.

A successful receipt requires all three facts:

- `Status::Ok`;
- exactly two accepted bytes; and
- chip select recorded inactive afterward.

No single fact substitutes for the others. A non-OK receipt may report zero or one accepted byte. Non-OK with two accepted bytes, or OK with a short byte count, is contradictory evidence and faults without advancing the generation.

Status	Accepted bytes	CS inactive	Disposition
OK	2	yes	accept the outstanding command
OK	0 or 1	either	contradictory; fault
non-OK	0 or 1	yes	transport failure with exact prefix
non-OK	2	either	contradictory; fault
either	any	no	cleanup evidence failed

5 Correlate before mutating

A receipt belongs to exactly one outstanding command. Its owner, lifecycle, configuration, presentation generation, operation index, register, and data must match. Observation time must be valid under modular ordering. An exact duplicate is idempotent; a changed duplicate, foreign receipt, stale receipt, regression, or exact-half-range timestamp rejects atomically.

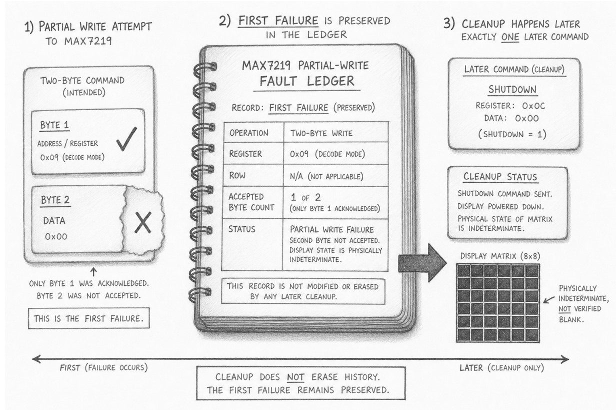
```
command = policy.service(noReceipt);
recorded = transport.record(command);
receipt = copyReceipt(command, recorded);
result = policy.service(receipt);
```

This pseudocode names the evidence flow; the canonical sketch contains the exact API. Before running it, predict which state may change after each line:

- **Command published:** one command is outstanding; no receipt has been accepted.
- **Transport recorded:** the policy is unchanged; the caller owns the copied trace.
- **Matching receipt accepted:** the operation may advance by exactly one.
- **Foreign receipt supplied:** no accepted state changes.

6 Make partial writes impossible to forget

A MAX7219 update is not physically atomic. If the first byte was accepted and the second failed, the controller cannot roll the display back. The policy therefore preserves desired state, last fully submitted state, accepted partial prefix, blank request, cleanup state, and physical indeterminacy as different facts.



Alternative text: a pencil incident ledger preserves the first failed operation, register, row, accepted-byte count, and status; a later one-command shutdown attempt is recorded in a separate cleanup box and never erases the primary entry.

After a failure, cleanup is pending. A later service call may request exactly one best-effort shutdown command. Its result is secondary evidence: cleanup status never overwrites the primary fault provenance. Even an accepted shutdown command proves only that the recording seam accepted its bytes. It does not prove that a physical matrix went dark.

> **Fault rule.** Preserve the first failure. Bound cleanup to a later single command. Label the physical presentation indeterminate until an electrically qualified endpoint provides stronger evidence.

7 Commit a complete frame, then submit rows

Frame candidates use the same transaction vocabulary as Lesson 058: `preview()`, `canCommit()`, and `commit()`. A preview is bound to owner, lifecycle, configuration, and base generation. Commit copies a complete desired frame but issues no transport command. Submitted generation advances only after all eight row receipts are accepted.

State	What it proves	What it does not prove
desired	a complete frame was committed locally	any command was accepted
partial prefix	some command prefix was accepted	a complete register write
last fully submitted	all eight row receipts passed	visible pixels or atomic update
shutdown accepted	shutdown bytes passed the seam	optical blanking
physically indeterminate	evidence cannot establish endpoint state	that the endpoint is lit or dark

The later Lesson 060 coordinator can preflight both display policies before either pure commit. That produces one coherent logical snapshot, not simultaneous photons.

8 Run the canonical replay

Open `examples/Lesson059Max7219Presentation/`
`Lesson059Max7219Presentation.ino`. Follow its narrative order: **acquire** the recording cells, **configure** orientation and intensity, **start** dark, then repeatedly **observe**, **decide**, and **present** one copied command.

Before running, record three predictions:

1. The first configuration register is ________.
2. Normal operation is requested before/after row clearing: ________.
3. A row-4 second-byte failure records accepted-byte count ________.

After running, inspect the named non-Serial result cells. Serial output may explain the trace, but it is not the only proof.

Observation cell	Expected result	Interpretation
configuration-order result	exact golden order accepted	policy starts dark and enables last
orientation result	asymmetric frame matches expected rows	transform is explicit
fault-provenance result	first failure and cleanup remain separate	no invented rollback
shutdown result	copied shutdown receipt recorded	logical request only

9 Diagnose evidence, not imagined hardware

Observed evidence	Likely policy meaning	Next check
no command emitted	receipt still outstanding or lifecycle inactive	inspect outstanding identity and lifecycle
receipt rejected without mutation	correlation, time, or structure mismatch	compare every copied field
cleanup pending	primary command failed or evidence contradicted itself	inspect first failure before cleanup
submitted generation unchanged	fewer than eight row receipts accepted	find the first incomplete row operation
orientation mismatch	transform expectation differs from configuration	replay one asymmetric pixel

10 Know the E0 boundary

This lesson proves deterministic frame transformation, exact register intent, one-command service, receipt correlation, partial-prefix provenance, bounded cleanup, lifecycle behavior, and compile-only Mega integration. It does not prove SPI electrical timing, module identity, RSET value, decoupling, matrix orientation, current, startup state, visible presentation, or safe blanking.

Before E1b promotion, record:

- exact MAX7219 IC or module revision and authoritative schematic;
- matrix anode/column and cathode/row mapping plus observed orientation;
- RSET and decoupling values, supply compatibility, and logic levels;
- serial timing, loaded-rail current, startup, fault, and shutdown evidence;
- a central resolution for the existing `SpiBus` terminal-fault recovery ambiguity before any MAX7219 adapter is added.

An unknown or low-value current-setting resistor may exceed the project's 100 mA USB campaign. Keep the module unpowered until its exact network is qualified.

11 Exit ticket

1. Why is a successful two-byte recording receipt not pixel evidence?
2. Why must an identical duplicate be idempotent but a changed duplicate reject?
3. Which state advances only after all eight row receipts pass?
4. Why is cleanup scheduled for a later service call?
5. Name two physical facts required before E1b promotion.

Completion evidence. You can predict the initialization order, transform an asymmetric frame, trace one command and matching receipt, explain every contradictory receipt, and distinguish a fully submitted logical frame from visible hardware state. Bench acceptance remains explicitly deferred.

Canonical companions: Lesson 059 sketch HTML lesson source implementation plan